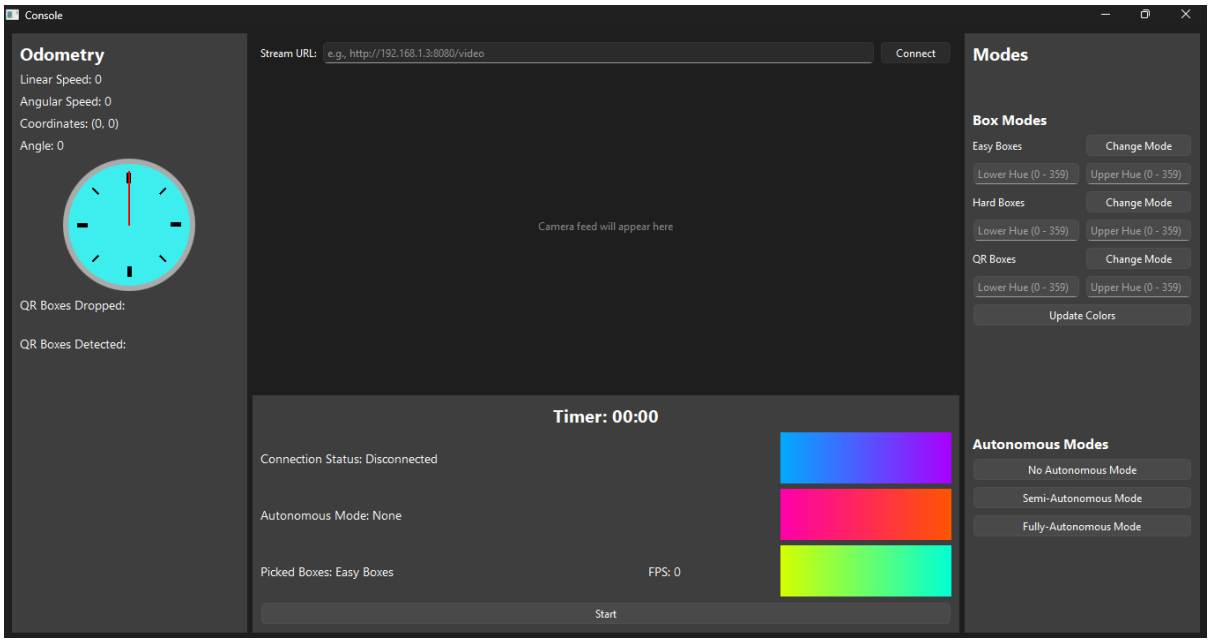
# Software Subteam Documentation

## 1.GUI




The console is implemented using PyQt6 and it handles data transfers using signals. It handles modes updates, odometry data display, camera feed, detection of QR codes, color changing of each type of boxes, and a communication channel with the ESP32.

### 1 - Camera Feed:

Camera feed is implemented using a VideoCaptureThread linked with IP Webcam mobile app. Another alternative was to transfer the camera feed through the ESP32, however that will create latency issues.

### 2 - Modes Bar:

Modes bar is used to change the autonomy modes, the types of picked boxes in an autonomous mode, and the colors of each type of boxes.

3 - Status Bar:

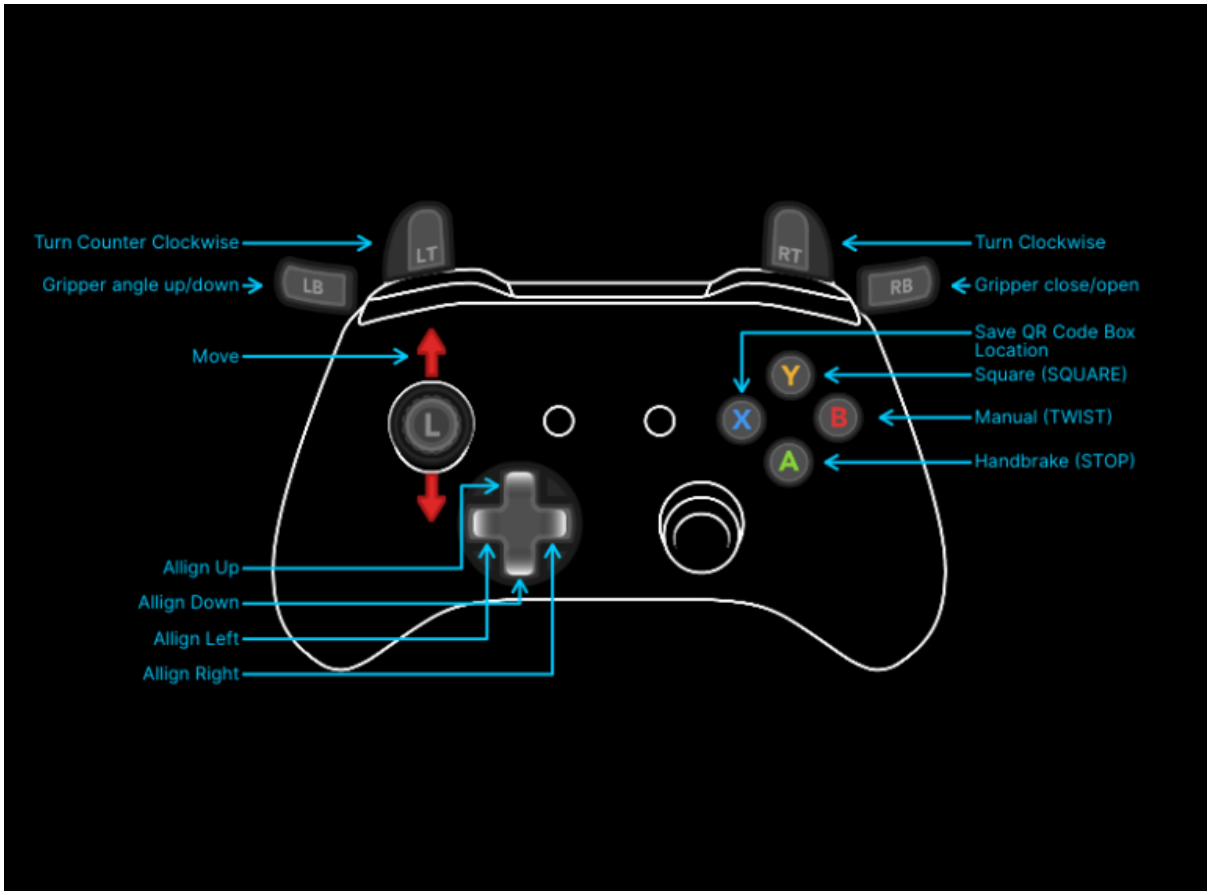
Status Bar is responsible for displaying the current connection status to camera feed, the current mode of autonomy and the gradients of each box color.

4 - Odometry Bar

Odometry Bar displays localization data, the location of the dropped QR code boxes, and the targeted location for the non-picked QR code boxes.

The console also handles QR Code reading using OpenCV and color detection of boxes [not used since robot autonomy has not been implemented].

The console supports both XBox controllers and PS controllers to drive the robot manually.



Communication Channel is implemented using a UDP Socket, and the console transmits and receives data with the ESP32 accordingly using the controllers' input.